

JESS Thermodynamic Database v8.9

Chemical species in reactions with Ruthenium

24-Jan-23

Charge	CAS	Count	Molecular formula	Mol. mass
Br-1 Bromide ion				
-1	24959-67-9	1213	Br (1)	79.9040
Caffeine Caffeine; 1,3,7-Trimethylxanthine; 3,7-Dihydro-1,3,7-trimethyl-1H-purine-2,6-dione; 1,3,7-Trimethyl-2,6-dioxopurine				
0	58-08-2	3	C (8) H (10) N (4) O (2)	194.193
Cl-1 Chloride ion				
-1	16887-00-6	2242	Cl (1)	35.4530
Cl2 (g) Chlorine gas				
0	7782-50-5	589	Cl (2)	70.9060
Cr+2 Chromium(II) ion				
2	22541-79-3	128	Cr (1)	51.9960
CrO4-2 Chromate ion				
-2	11104-59-9	127	Cr (1) O (4)	115.994
e-1 Electron				
-1	---	3399	E (1)	0.000000
EDTA-4 Ethylenediaminetetraacetate anion				
-4	---	1457	C (10) H (12) N (2) O (8)	288.214

F2 (g) Fluorine gas				
0	7782-41-4	259	F (2)	37.9968
H+1 Hydrogen ion; Proton				
1	12408-02-5	31679	H (1)	1.00794
H+1_RuO4-1				
0	---	1	H (1) O (4) Ru (1)	166.078
H2 (g) Hydrogen gas; Hydrogen				
0	1333-74-0	1564	H (2)	2.01588
H2O Water				
0	7732-18-5	5947	H (2) O (1)	18.0153
I-1 Iodide ion				
-1	20461-54-5	894	I (1)	126.900
Ir+4_Cl-1 (5)				
-1	---	7	Cl (5) Ir (1)	369.485
Ir+4_Ru+4_Cl-1 (10)_OH-1				
-3	---	1	Cl (10) H (1) Ir (1) O (1) Ru (1)	664.829
Ir+4_Ru+4_Cl-1 (6)_OH-1 (5)				
-3	---	1	Cl (6) H (5) Ir (1) O (5) Ru (1)	591.047
Ir+4_Ru+4_Cl-1 (7)_OH-1 (4)				
-3	---	1	Cl (7) H (4) Ir (1) O (4) Ru (1)	609.492
Ir+4_Ru+4_Cl-1 (8)_OH-1 (3)				
-3	---	1	Cl (8) H (3) Ir (1) O (3) Ru (1)	627.938

Ir+4_Ru+4_Cl-1(9)_OH-1(2)				
-3	---	1	Cl(9)H(2)Ir(1)O(2)Ru(1)	646.384
K+1 Potassium(I) ion				
1	24203-36-9	462	K(1)	39.0980
MnO4-1 Permanganate ion				
-1	14333-13-2	20	Mn(1)O(4)	118.936
MnO4-2 Manganate ion				
-2	39321-12-5	19	Mn(1)O(4)	118.936
N2 Nitrogen (aquated)				
0	---	3	N(2)	28.0134
N2O Nitrous oxide (aquated)				
0	---	3	N(2)O(1)	44.0128
NO Nitric oxide (aquated)				
0	10102-43-9	16	N(1)O(1)	30.0061
NO3-1 Nitrate ion				
-1	14797-55-8	573	N(1)O(3)	62.0049
O2 (g) Oxygen gas; Oxygen				
0	7782-44-7	2559	O(2)	31.9988
OH-1 Hydroxide ion				
-1	14280-30-9	6558	H(1)O(1)	17.0073

PhenThiourea Phenylthiourea; Phenylthiocarbamide; 1-Phenyl-2-thiourea				
0	103-85-5	21	C(7)H(8)N(2)S(1)	152.214
Ru+2 Ruthenium(II) ion				
2	22541-59-9	24	Ru(1)	101.072
Ru+2_Bipy(3)				
2	---	1	C(30)H(24)N(6)Ru(1)	569.633
Ru+2_Caffeine				
2	---	1	C(8)H(10)N(4)O(2)Ru(1)	295.265
Ru+2_Cl-1				
1	---	1	Cl(1)Ru(1)	136.525
Ru+2_Cl-1(2)				
0	---	1	Cl(2)Ru(1)	171.978
Ru+2_Cl-1(3)				
-1	---	1	Cl(3)Ru(1)	207.431
Ru+2_Cl-1(6)				
-4	---	1	Cl(6)Ru(1)	313.790
Ru+2_CN-1(6)				
-4	---	2	C(6)N(6)Ru(1)	257.178
Ru+2_CrO4-2				
0	---	1	Cr(1)O(4)Ru(1)	217.066
Ru+2_EtDiAm(3)				
2	---	1	C(6)H(24)N(6)Ru(1)	281.369
Ru+2_H+1_Pyrazine_CN-1(5)				
-2	---	1	C(9)H(5)N(7)Ru(1)	312.258

Ru+2_H+1(-1)_Imidazole_NH3(5)				
1	---	1	C(3)H(18)N(7)Ru(1)	253.295
Ru+2_H+1(-1)_NH3(5)_His-1				
0	---	1	C(6)H(22)N(8)O(2)Ru(1)	339.365
Ru+2_Imidazole_NH3(5)				
2	---	1	C(3)H(19)N(7)Ru(1)	254.303
Ru+2_K+1_CN-1(6)				
-3	---	1	C(6)K(1)N(6)Ru(1)	296.276
Ru+2_N2_NH3(5)				
2	---	2	H(15)N(7)Ru(1)	214.238
Ru+2_N2O_NH3(5)				
2	---	1	H(15)N(7)O(1)Ru(1)	230.237
Ru+2_NH3(5)				
2	---	5	H(15)N(5)Ru(1)	186.225
Ru+2_NH3(5)_Br-1				
1	---	1	Br(1)H(15)N(5)Ru(1)	266.129
Ru+2_NH3(5)_Cl-1				
1	---	1	Cl(1)H(15)N(5)Ru(1)	221.678
Ru+2_NH3(5)_His-1				
1	---	1	C(6)H(23)N(8)O(2)Ru(1)	340.373
Ru+2_NH3(6)				
2	---	1	H(18)N(6)Ru(1)	203.255
Ru+2_Pyrazine_CN-1(5)				
-3	---	1	C(9)H(4)N(7)Ru(1)	311.250

Ru+2_SO4-2				
0	---	1	O (4) Ru (1) S (1)	197.130
Ru+2_Theophylline-1				
1	---	1	C (7) H (7) N (4) O (2) Ru (1)	280.230
Ru+2 (2)_N2_NH3 (10)				
4	---	1	H (30) N (12) Ru (2)	400.463
Ru+3 Ruthenium(III) ion				
3	22541-88-4	45	Ru (1)	101.072
Ru+3_Bipy (3)				
3	---	1	C (30) H (24) N (6) Ru (1)	569.633
Ru+3_Caffeine				
3	---	1	C (8) H (10) N (4) O (2) Ru (1)	295.265
Ru+3_Cl-1				
2	---	3	Cl (1) Ru (1)	136.525
Ru+3_Cl-1 (2)				
1	---	4	Cl (2) Ru (1)	171.978
Ru+3_Cl-1 (3)				
0	---	3	Cl (3) Ru (1)	207.431
Ru+3_Cl-1 (3)_(s) Ruthenium trichloride				
0	---	2	Cl (3) Ru (1)	207.431
Ru+3_Cl-1 (4)				
-1	---	3	Cl (4) Ru (1)	242.884
Ru+3_Cl-1 (5)				
-2	---	2	Cl (5) Ru (1)	278.337

Ru+3_Cl-1 (6)				
-3	---	3	Cl (6) Ru (1)	313.790
Ru+3_CN-1 (6)				
-3	---	1	C (6) N (6) Ru (1)	257.178
Ru+3_Cr+2_Cl-1				
4	---	1	Cl (1) Cr (1) Ru (1)	188.521
Ru+3_EDTA-4				
-1	---	1	C (10) H (12) N (2) O (8) Ru (1)	389.286
Ru+3_EtDiAm (3)				
3	---	1	C (6) H (24) N (6) Ru (1)	281.369
Ru+3_NH3 (5)				
3	---	3	H (15) N (5) Ru (1)	186.225
Ru+3_NH3 (5)_Br-1				
2	---	1	Br (1) H (15) N (5) Ru (1)	266.129
Ru+3_NH3 (5)_Cl-1				
2	---	1	Cl (1) H (15) N (5) Ru (1)	221.678
Ru+3_NH3 (5)_I-1				
2	---	1	H (15) I (1) N (5) Ru (1)	313.125
Ru+3_NH3 (6)				
3	---	1	H (18) N (6) Ru (1)	203.255
Ru+3_NO				
3	---	5	N (1) O (1) Ru (1)	131.078
Ru+3_NO_Cl-1 (3)				
0	34416-98-3	1	Cl (3) N (1) O (1) Ru (1)	237.437

Ru+3_NO_Cl-1(3)_OH-1				
-1	---	2	Cl(3)H(1)N(1)O(2)Ru(1)	254.444
Ru+3_NO_Cl-1(3)_OH-1(2)				
-2	---	1	Cl(3)H(2)N(1)O(3)Ru(1)	271.452
Ru+3_NO_Cl-1(4)				
-1	68297-36-9	2	Cl(4)N(1)O(1)Ru(1)	272.890
Ru+3_NO_Cl-1(4)_H2O				
-1	---	1	Cl(4)H(2)N(1)O(2)Ru(1)	290.905
Ru+3_NO_Cl-1(4)_OH-1				
-2	---	2	Cl(4)H(1)N(1)O(2)Ru(1)	289.897
Ru+3_NO_NO3-1				
2	---	1	N(2)O(4)Ru(1)	193.083
Ru+3_NO_NO3-1(2)				
1	---	1	N(3)O(7)Ru(1)	255.088
Ru+3_NO_NO3-1(3)				
0	---	1	N(4)O(10)Ru(1)	317.093
Ru+3_OH-1(3)_(s) Ruthenium(III) hydroxide				
0	12135-42-1	2	H(3)O(3)Ru(1)	152.094
Ru+3_PhenThiourea_Cl-1(4)				
-1	---	1	C(7)H(8)Cl(4)N(2)Ru(1)S(1)	395.098
Ru+3_PhenThiourea(2)_Cl-1(3)				
0	---	1	C(14)H(16)Cl(3)N(4)Ru(1)S(2)	511.859
Ru+3_PhenThiourea(3)_Cl-1(2)				
1	---	1	C(21)H(24)Cl(2)N(6)Ru(1)S(3)	628.620

Ru+3_SCN-1				
2	---	1	C (1) N (1) Ru (1) S (1)	159.150
Ru+3_SO4-2				
1	---	2	O (4) Ru (1) S (1)	197.130
Ru+3_SO4-2 (2)				
-1	---	2	O (8) Ru (1) S (2)	293.187
Ru+3_Theophylline-1				
2	---	1	C (7) H (7) N (4) O (2) Ru (1)	280.230
Ru+3_Thiourea_Cl-1 (4)				
-1	---	1	C (1) H (4) Cl (4) N (2) Ru (1) S (1)	319.000
Ru+3_Thiourea (2)_Cl-1 (3)				
0	---	1	C (2) H (8) Cl (3) N (4) Ru (1) S (2)	359.663
Ru+3_Thiourea (3)_Cl-1 (2)				
1	---	1	C (3) H (12) Cl (2) N (6) Ru (1) S (3)	400.326
Ru+4 Ruthenium(IV) ion				
4	22541-58-8	35	Ru (1)	101.072
Ru+4_Cl-1				
3	---	2	Cl (1) Ru (1)	136.525
Ru+4_Cl-1_OH-1 (2)				
1	---	1	Cl (1) H (2) O (2) Ru (1)	170.540
Ru+4_Cl-1_OH-1 (5)				
-2	---	1	Cl (1) H (5) O (5) Ru (1)	221.562
Ru+4_Cl-1 (2)				
2	---	2	Cl (2) Ru (1)	171.978

Ru+4 _Cl-1 (2) _OH-1 (2)				
0	---	1	Cl (2) H (2) O (2) Ru (1)	205.993
Ru+4 _Cl-1 (2) _OH-1 (4)				
-2	---	1	Cl (2) H (4) O (4) Ru (1)	240.007
Ru+4 _Cl-1 (3)				
1	---	2	Cl (3) Ru (1)	207.431
Ru+4 _Cl-1 (3) _OH-1 (2)				
-1	---	2	Cl (3) H (2) O (2) Ru (1)	241.446
Ru+4 _Cl-1 (3) _OH-1 (3)				
-2	---	1	Cl (3) H (3) O (3) Ru (1)	258.453
Ru+4 _Cl-1 (4)				
0	---	2	Cl (4) Ru (1)	242.884
Ru+4 _Cl-1 (4) _OH-1 (2)				
-2	---	4	Cl (4) H (2) O (2) Ru (1)	276.899
Ru+4 _Cl-1 (5)				
-1	---	2	Cl (5) Ru (1)	278.337
Ru+4 _Cl-1 (5) _OH-1				
-2	---	3	Cl (5) H (1) O (1) Ru (1)	295.344
Ru+4 _Cl-1 (6)				
-2	---	1	Cl (6) Ru (1)	313.790
Ru+4 _CrO4-2				
2	---	1	Cr (1) O (4) Ru (1)	217.066
Ru+4 _OH-1 (2)				
2	---	7	H (2) O (2) Ru (1)	135.087

Ru+4_OH-1(4)_(s) Ruthenium(IV) hydroxide				
0	---	1	H(4)O(4)Ru(1)	169.101
Ru+4_S-2(2)_(s) Ruthenium disulfide				
0	---	3	Ru(1)S(2)	165.192
Ru+4_Se-2(2)_(s) Ruthenium diselenide				
0	---	1	Ru(1)Se(2)	258.998
Ru(g)				
0	---	1	Ru(1)	101.072
Ru(s) Ruthenium				
0	7440-18-8	18	Ru(1)	101.072
RuF5(s) Ruthenium pentafluoride				
0	---	1	F(5)Ru(1)	196.064
RuO2.2H2O(s) Ruthenium dioxide dihydrate; Ruthenium oxide				
0	80498-27-7	3	H(4)O(4)Ru(1)	169.101
RuO2(am.,s) Ruthenium dioxide, amorphous				
0	---	1	O(2)Ru(1)	133.071
RuO2(s) Ruthenium(IV) oxide; Ruthenium dioxide				
0	12036-10-1	4	O(2)Ru(1)	133.071
RuO4 Ruthenium(VIII) oxide aqueous				
0	---	7	O(4)Ru(1)	165.070
RuO4_H2O				

0	---	1	H (2) O (5) Ru (1)	183.085
RuO4_OH-1				
-1	---	2	H (1) O (5) Ru (1)	182.077
RuO4-1				
-1	---	9	O (4) Ru (1)	165.070
RuO4-2				
-2	---	6	O (4) Ru (1)	165.070
RuO4 (g) Ruthenium tetroxide gas				
0	---	1	O (4) Ru (1)	165.070
RuO4 (OH) -1 Ruthenium Oxide Hydroxide ion				
-1	---	1	H (1) O (5) Ru (1)	182.077
RuO4 (s) Ruthenium(VIII) oxide; Ruthenium(VIII) tetroxide; Ruthenium tetroxide				
0	20427-56-9	7	O (4) Ru (1)	165.070
S-2 Sulfide ion; Sulphide ion				
-2	14127-58-3	466	S (1)	32.0600
S (s) Sulphur, alpha; Sulphur, orthorhombic; Sulphur; Sulfur, alpha; Sulfur, orthorhombic; Sulfur; Sulfur-Rhmb				
0	7704-34-9	619	S (1)	32.0600
SCN-1 Thiocyanate ion; NCS-1 equivalent				
-1	302-04-5	785	C (1) N (1) S (1)	58.0777
Se (s) Selenium, trigonal; Selenium, grey; Selenium, black; Selenium, gamma; Selenium, hexagonal (sic)				
0	7782-49-2	140	Se (1)	78.9630

SO₄-2 Sulfate ion; Sulphate ion				
-2	14808-79-8	1270	O(4)S(1)	96.0576
Theophylline-1				
-1	---	3	C(7)H(7)N(4)O(2)	179.158
Thiourea Thiocarbamide; Thiourea; tu				
0	62-56-6	261	C(1)H(4)N(2)S(1)	76.1162